Calibration-Aware Recurrent Neural Networks for Multivariate Probabilistic Forecasting under Distribution Shift

Jonathan Ma

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Abstract

Probabilistic forecasts are useful only when their stated uncertainty is credible. Likelihood-trained recurrent neural networks may be miscalibrated, while Bayesian model averaging is not guaranteed to remain calibrated when the likelihood, prior, or conditional dynamics are misspecified. This paper develops the Calibration-Aware Recurrent Neural Network (CA-RNN), adapting the calibration-aware frequentist and Bayesian learning framework of Huang, Park, and Simeone to continuous multivariate forecasting. The proposed training criterion augments joint negative log likelihood with a differentiable discrepancy between projected probability integral transforms (PITs) and the uniform distribution. A sequential extension additionally penalizes lagged dependence in the projected PIT sequence. CA-RNN is a conditional stochastic forecasting model: its recurrent state parameterizes a predictive distribution from which future observations can be sampled. It is not presented as a latent-variable or likelihood-free deep generative architecture such as a diffusion model, normalizing flow, generative adversarial network, or variational autoencoder. The empirical design compares RNN, Bayesian RNN, CA-RNN, and calibration-aware Bayesian RNN, and then extends the comparison to model misspecification and distribution shift. A frozen replication on a disjoint four-asset panel confirms a smaller projected-PIT calibration improvement, but finds significantly worse MSE, Energy Score, interval score, and PIT temporal dependence than the likelihood-trained RNN. A VAR with marginal GARCH$(1,1)$ scales and joint standardized-residual resampling provides substantially better interval calibration than either neural model. The results show that directly optimizing a calibration diagnostic can generalize to new assets without improving the predictive distribution as judged by proper scores. Supporting results prove that the finite calibration criterion is non-identifying, construct dependent uniform PITs with zero covariance, and express Energy and interval-score regret in quantities that the calibration loss does not control.

Contents

1 Introduction

1.1 Motivation

A point forecast reports one possible future; a probabilistic forecast reports a distribution of possible futures. The latter is operationally meaningful only if events assigned probability α occur at approximately rate α, while the distribution remains sufficiently sharp to discriminate among outcomes. Modern neural forecasters can optimize likelihood successfully without achieving reliable finite-sample calibration. Bayesian learning represents parameter uncertainty, but its calibration guarantees depend on idealized assumptions about the likelihood, prior, posterior computation, and data-generating mechanism.

Throughout this paper, CA-RNN is described precisely as a *calibration-aware conditional stochastic forecasting model*. It is probabilistic because it estimates a full conditional law rather than only a conditional mean, stochastic because forecasts are obtained by drawing innovations from that law, and generative in the broad operational sense that it can sample future observations and trajectories. We do not use "deep generative model" as a model-class label: the present architecture has no learned latent diffusion, invertible flow, adversarial generator, or variational latent state. Its novelty lies in calibration-aware estimation of an RNN-indexed predictive distribution.

Huang, Park, and Simeone propose calibration-aware Bayesian neural networks (CA-BNNs), combining a data-dependent calibration regularizer with the data-independent KL regularizer of variational Bayesian learning [1]. Their experiments compare frequentist, Bayesian, calibration-aware frequentist, and calibration-aware Bayesian classifiers; vary the calibration weight; and report expected calibration error and reliability diagrams. Their conclusion identifies evaluation under distribution shift as a direction for future work. This paper transfers that methodological template to continuous multivariate sequential prediction and makes misspecification and distribution shift a central part of the evaluation.

1.2 Research questions

The primary research question is:

> How does augmenting a likelihood-trained Gaussian RNN with differentiable projected-PIT calibration penalties affect the calibration and predictive quality of one-step-ahead multivariate financial forecasts, both across chronological market regimes and relative to likelihood-only neural and VAR–GARCH benchmarks?

This question is decomposed into three focused empirical questions:

RQ1. Across the prespecified calibration-weight grid, what tradeoff arises between projected calibration error and strictly proper predictive scores, marginal interval score, and point-forecast error?

RQ2. Conditional on the marginal projected-PIT penalty, does adding the lagged PIT-product penalty reduce held-out sequential calibration error, and how does it affect Energy Score, marginal interval score, coverage, and point-forecast error relative to marginal CA–RNN and the likelihood-only RNN?

RQ3. Does sequential CA–RNN add forecast value beyond the likelihood-only RNN and VAR–GARCH on the development and frozen external asset panels, and how does its performance relative to the RNN vary across four chronological market regimes, as measured by projected and sequential calibration, Energy Score, interval score, coverage, and point-forecast error?

1.3 Contributions

The contributions are methodological, theoretical, and empirical:

1. a continuous-outcome analogue of calibration-aware neural training based on differentiable projected PIT discrepancies;
2. an extension targeting temporal dependence in forecast PIT sequences;
3. a variational Bayesian recurrent formulation combining calibration and KL regularization;
4. a matched four-model experiment that parallels [1]; and
5. a controlled study of calibration degradation under misspecification and distribution shift, including a multivariate financial application;
6. comparison with a volatility-aware traditional benchmark; and
7. a frozen external-panel replication separating developmental findings from confirmatory evidence; and
8. a theoretical account of the negative confirmatory result through the failure of finite moments to identify the distribution, a PIT-dependence counterexample, and exact proper-score regret representations.

1.4 Scope and non-claims

Finite projected calibration is weaker than complete multivariate distributional calibration. Calibration of a finite test set is also not a guarantee of conditional calibration at every history. We therefore avoid claims that a small collection of projections proves joint correctness. The proposed regularizer is an explicit, diagnostically aligned training objective whose value must be established empirically with proper scores and out-of-sample tests.

2 Related Work

2.1 Calibration-aware Bayesian learning

The organizing reference is [1]. It distinguishes data-dependent calibration regularization from the data-independent prior regularization used by variational Bayesian inference. Its CA-FNN objective adds a differentiable approximation of expected calibration error to cross-entropy, and its CA-BNN objective places the same calibration penalty inside the variational expectation while retaining a KL penalty. The paper uses Gaussian variational parameters and the reparameterization trick, and its main comparison comprises FNN, BNN, CA-FNN, and CA-BNN. The reported experiments emphasize calibration-weight sweeps and reliability diagrams.

2.2 Calibration, sharpness, and proper scoring

Probabilistic calibration concerns the statistical consistency between forecast distributions and realized outcomes. Gneiting, Balabdaoui, and Raftery [2] organize forecast evaluation around the principle that a predictive distribution should be as sharp as possible subject to calibration. This distinction is central here: a diffuse predictive law can achieve acceptable empirical coverage without being informative, while a sharp but systematically incorrect law is not credible. Calibration diagnostics must therefore be accompanied by measures that reward the entire predictive distribution.

Proper scoring rules provide that complement. A scoring rule is proper when its expected value is optimized by reporting the data-generating distribution and is strictly proper when that optimum is unique [3]. Negative log predictive density, the energy score, and the interval score consequently answer a different question from a finite calibration penalty: they assess the quality of the reported distribution rather than agreement with selected moment conditions. For multivariate quantities, the energy score extends the continuous ranked probability score and supports sample-based evaluation [4]. It can, however, have limited sensitivity to misspecified dependence; variogram scores were developed in part to expose such errors [5]. The manuscript therefore reports proper scores alongside projected calibration, marginal intervals, and portfolio-tail risk rather than interpreting any one diagnostic as sufficient.

More recent work formalizes conditional calibration and links forecast evaluation to regression diagnostics [6]. This is an important qualification for the present method. CA–RNN enforces a finite set of unconditional empirical restrictions on projected PITs; it does not establish auto-calibration conditional on every forecast history. The distinction motivates both the non-identification results in Section 4.6 and the use of genuinely out-of-sample proper-score comparisons.

2.3 Density-forecast diagnostics under temporal dependence

Probability integral transforms (PITs) convert a correctly specified continuous univariate predictive distribution into uniform observations. Diebold, Gunther, and Tay [7] made PIT histograms and dependence diagnostics a practical framework for evaluating financial density forecasts. Berkowitz [8] proposed likelihood-based tests after a normal transformation of the PIT, allowing uniformity and independence to be examined jointly. For overlapping or multi-step forecasts, however, transformed observations may be serially correlated even under the null. Knüppel [9] develops raw-moment tests that explicitly accommodate this dependence.

These results explain the design of the CA–RNN penalty. Marginal projected PIT moments target distributional shape, while lagged PIT-product moments target one limited form of sequential dependence. The latter is diagnostically relevant, but neither zero sample autocovariance nor uniformity on a finite grid proves independence. Consequently, sequential calibration error is treated as a targeted diagnostic, not as a complete specification test.

2.4 Calibration-aware neural prediction

Modern neural networks can produce poorly calibrated probabilities even when their predictive accuracy is high. Guo et al. [10] document this phenomenon in classification and show the effectiveness of post-hoc temperature scaling. Kumar, Sarawagi, and Jain [11] instead construct a differentiable kernel-based calibration measure that can be included during training. For continuous outcomes, Kuleshov, Fenner, and Ermon [12] propose a post-hoc recalibration procedure for predictive cumulative distribution functions and demonstrate it with deep regression and recurrent models. These papers establish that calibration can be measured or corrected in neural systems, but they also illustrate the distinction between post-hoc marginal recalibration and joint, sequential calibration-aware estimation.

Huang, Park, and Simeone [1] bring calibration directly into Bayesian neural learning. Their calibration-aware objective combines a data-dependent differentiable calibration loss with a data-independent variational KL regularizer. CA–RNN preserves this organizing idea but changes the statistical object: class probabilities are replaced by multivariate one-step-ahead predictive distributions, and calibration is evaluated through projected PIT and sequential moment restrictions. This adaptation is not mechanical. Forecast observations are ordered and dependent, predictive covariance matters, and a multivariate law cannot be validated by marginal coverage alone.

2.5 Probabilistic recurrent forecasting, misspecification, and shift

DeepAR demonstrates that autoregressive recurrent networks can estimate conditional predictive distributions by likelihood and share information across related series [13]. CA–RNN adopts the recurrent conditional-density perspective but uses a joint multivariate Gaussian head and a variational parameter distribution, then augments likelihood-based estimation with calibration restrictions. This separation is consequential under misspecification. White [14] shows that maximum likelihood estimation under a misspecified family generally converges to a pseudo-true distribution rather than the data-generating law. Calibration-aware regularization can move the fitted distribution away from that likelihood optimum, but such movement need not improve a strictly proper score. The exact regret arguments developed below formalize this tradeoff.

Distribution shift creates a second challenge. Ovadia et al. [15] show empirically that predictive uncertainty methods that appear adequate in distribution can deteriorate as dataset shift intensifies. Financial returns add pronounced conditional heteroskedasticity; the GARCH model of Bollerslev [16] provides the canonical likelihood-based mechanism for persistent time-varying variance. These strands motivate the manuscript's shifted-regime experiments and its VAR–GARCH benchmark. A neural calibration method should not receive credit merely for learning volatility that a standard econometric model already represents directly.

2.6 Positioning of the present study

Taken together, the literature supplies mature tools for proper forecast evaluation, PIT-based density diagnostics, neural recalibration, recurrent probabilistic prediction, and uncertainty evaluation under shift. The present study combines these tools to ask a narrower unresolved question: whether an in-training calibration penalty for a Bayesian recurrent forecaster improves joint probabilistic forecasts when the model is misspecified or the data regime changes. Its contribution is therefore methodological and diagnostic rather than a claim that finite projected calibration identifies the true multivariate law. The confirmatory design deliberately tests whether improvement in the optimized projected criterion transfers to proper scores, interval quality, tail-risk estimates, and an external asset panel. The observed failure of that transfer is itself informative about the limits of calibration-aware objectives for dependent continuous outcomes.

3 Forecasting Setup and Calibration

3.1 Sequential prediction problem

Let $(Y_t)_{t=1}^T$ be an adapted stochastic process with $Y_t \in \mathbb{R}^d$ and filtration $\mathcal{F}_t = \sigma(Y_1, \ldots, Y_t)$. Given a history length H, define $X_t = (Y_{t-H+1}, \ldots, Y_t)$. The task is to estimate a conditional predictive distribution

$$P_t(\cdot) = \mathbb{P}(Y_{t+1} \in \cdot \mid \mathcal{F}_t)$$

with a model $P_{\theta,t}$. Training, validation, and test blocks are ordered chronologically. Standardization parameters are estimated on the training block only.

3.2 Joint Gaussian predictive head

For the initial specification, the predictive law is

$$Y_{t+1} \mid \mathcal{F}_t, \theta \sim \mathcal{N}_d(\mu_{\theta,t}, \Sigma_{\theta,t}), \qquad \Sigma_{\theta,t} = L_{\theta,t} L_{\theta,t}^{\mathsf{T}}. \tag{1}$$

The network outputs the d components of $\mu_{\theta,t}$ and the $d(d+1)/2$ lower-triangular entries of $L_{\theta,t}$. An unconstrained diagonal output $s_{t,j}$ is transformed as

$$L_{t,jj} = \mathrm{softplus}(s_{t,j}) + \varepsilon, \qquad \varepsilon > 0.$$

Proposition 3.1 (Validity of the predictive covariance)**.** *For every network output, $\Sigma_{\theta,t} = L_{\theta,t} L_{\theta,t}^{\mathsf{T}}$ is symmetric positive definite.*

Proof. Symmetry follows from transposition. Since every diagonal entry of the triangular matrix $L_{\theta,t}$ is strictly positive, $L_{\theta,t}$ is nonsingular. Hence, for any nonzero $x \in \mathbb{R}^d$,

$$x^{\mathsf{T}} \Sigma_{\theta,t} x = x^{\mathsf{T}} L_{\theta,t} L_{\theta,t}^{\mathsf{T}} x = \|L_{\theta,t}^{\mathsf{T}} x\|_2^2 > 0.$$

Thus $\Sigma_{\theta,t}$ is positive definite. □

3.3 Negative log likelihood

For an index set I, the mean joint Gaussian negative log likelihood is

$$\mathcal{L}_{\mathrm{NLL}}(\theta; I) = -\frac{1}{|I|} \sum_{t \in I} \log p_\theta(Y_{t+1} \mid X_t) \tag{2}$$

$$= \frac{1}{2|I|} \sum_{t \in I} \left[d \log(2\pi) + 2 \sum_{j=1}^{d} \log L_{t,jj} + e_t^{\mathsf{T}} \Sigma_{\theta,t}^{-1} e_t \right], \tag{3}$$

where $e_t = Y_{t+1} - \mu_{\theta,t}$. Computation uses triangular solves rather than an explicit matrix inverse.

Proposition 3.2 (Strict propriety of the population log score)**.** *Let $P_0(\cdot \mid \mathcal{F}_t)$ have density p_0 and let p be any candidate conditional density with common support. Then*

$$\mathbb{E}_{P_0}[-\log p(Y_{t+1} \mid \mathcal{F}_t)] - \mathbb{E}_{P_0}[-\log p_0(Y_{t+1} \mid \mathcal{F}_t)] = \mathbb{E}[\mathrm{KL}(P_0(\cdot \mid \mathcal{F}_t) \| P(\cdot \mid \mathcal{F}_t))] \geq 0.$$

Equality holds only when the conditional densities agree almost surely.

Proof. Condition on $\mathcal{F}_t$, subtract the two expected log scores, and recognize the conditional KL divergence. Nonnegativity follows from Gibbs' inequality. Taking expectations over $\mathcal{F}_t$ proves the result. □

This proposition explains why the calibration penalty must supplement rather than replace a proper score: calibration alone can admit uninformative forecasts.

3.4 Projected probability integral transforms

Fix unit vectors $a_1, \ldots, a_R \in \mathbb{R}^d$. In the experiments they comprise the d coordinate directions, one equal-weight direction, and four random unit directions fixed by a documented seed. Define

$$U_{t,r} = F_{\theta,t,r}(a_r^\top Y_{t+1}), \tag{4}$$

where $F_{\theta,t,r}$ is the predictive CDF of $a_r^\top Y_{t+1}$.

Proposition 3.3 (Projected Gaussian law)**.** *Under* (1),

$$a_r^\top Y_{t+1} \mid \mathcal{F}_t, \theta \sim \mathcal{N}\left(a_r^\top \mu_{\theta,t}, a_r^\top \Sigma_{\theta,t} a_r\right),$$

and therefore

$$U_{t,r} = \Phi\left(\frac{a_r^\top (Y_{t+1} - \mu_{\theta,t})}{\sqrt{a_r^\top \Sigma_{\theta,t} a_r}}\right).$$

Proof. A linear transformation of a multivariate Gaussian is Gaussian. Its mean and variance follow from linearity of expectation and the covariance transformation rule. Applying its CDF gives the displayed PIT. □

Theorem 3.4 (Sequential PIT property)**.** *Suppose $F_{0,t,r}$ is the true, continuous conditional CDF of $a_r^\top Y_{t+1}$ given $\mathcal{F}_t$. Then $U_{t,r} = F_{0,t,r}(a_r^\top Y_{t+1})$ is uniformly distributed conditional on $\mathcal{F}_t$. For a fixed r, the sequence $(U_{t,r})_t$ is independent and* Uniform$(0,1)$.

Proof. For $u \in [0,1]$, continuity and the conditional probability integral transform give

$$\mathbb{P}(U_{t,r} \le u \mid \mathcal{F}_t) = u.$$

Thus $U_{t,r}$ is conditionally uniform and its conditional law does not depend on $\mathcal{F}_t$. For times $t_1 < \cdots < t_m$, iterated conditioning yields

$$\begin{aligned}\mathbb{P}(U_{t_1,r} \le u_1, \ldots, U_{t_m,r} \le u_m) &= \mathbb{E}\left[\mathbf{1}_{\{U_{t_1,r} \le u_1, \ldots, U_{t_{m-1},r} \le u_{m-1}\}} \mathbb{P}(U_{t_m,r} \le u_m \mid \mathcal{F}_{t_m})\right] \\ &= u_m \mathbb{P}(U_{t_1,r} \le u_1, \ldots, U_{t_{m-1},r} \le u_{m-1}).\end{aligned}$$

Induction gives $\prod_{j=1}^m u_j$, proving independence and uniformity. □

Remark 3.5. *Uniformity for finitely many a_r is necessary but not sufficient for equality of multivariate distributions. If all one-dimensional projections agree, the Cramér–Wold device identifies the joint distribution; a finite projection set only provides a tractable approximation and diagnostic.*

Training uses the analytic Gaussian PIT above. Out-of-sample evaluation instead uses the common sample-rank approximation

$$\widehat{U}_{t,r} = \frac{\sum_{s=1}^S \mathbf{1}\{a_r^\top Y_{t+1}^{(s)} < a_r^\top Y_{t+1}\} + 1/2}{S+1},$$

with $S = 500$ predictive draws. This supports the same diagnostic for deterministic Gaussian forecasts and Bayesian posterior-predictive mixtures, at the cost of finite-simulation error.

4 Calibration-Aware RNN Methodology

4.1 Recurrent architecture

For a single-layer GRU, with elementwise product $\odot$,

$$\begin{aligned} r_t &= \sigma(W_{ir}Y_t + b_{ir} + W_{hr}h_{t-1} + b_{hr}), \\ z_t &= \sigma(W_{iz}Y_t + b_{iz} + W_{hz}h_{t-1} + b_{hz}), \\ n_t &= \tanh(W_{in}Y_t + b_{in} + r_t \odot (W_{hn}h_{t-1} + b_{hn})), \\ h_t &= n_t + z_t \odot (h_{t-1} - n_t). \end{aligned}$$

Linear heads map the final hidden state to $\mu_{\theta,t}$ and the unconstrained entries of $L_{\theta,t}$.

4.2 Differentiable projected calibration loss

Let $q_g = g/(G+1)$ for $g = 1, \ldots, G$. The empirical CDF of PIT values is nondifferentiable in model parameters because it uses indicators. Replace the indicator $\mathbf{1}\{U_{t,r} \le q_g\}$ by

$$s_\tau(q_g - U_{t,r}) = \sigma\left(\frac{q_g - U_{t,r}}{\tau}\right), \qquad \tau > 0.$$

For a consecutive training batch B, define

$$\mathcal{L}_{\text{cal}}(\theta; B) = \frac{1}{GR}\sum_{g=1}^{G}\sum_{r=1}^{R}\left[\frac{1}{|B|}\sum_{t\in B} s_\tau(q_g - U_{t,r}) - q_g\right]^2. \tag{5}$$

This is a grid approximation to an integrated squared CDF discrepancy, analogous in purpose—but not in definition—to the differentiable AECE used for classification in [1].

Proposition 4.1 (Differentiability)**.** *If the network outputs are differentiable in θ and $a_r^\top \Sigma_{\theta,t} a_r > 0$, then $\mathcal{L}_{\text{cal}}(\theta; B)$ is differentiable in θ.*

Proof. The Cholesky parameterization, affine projection, positive square root, Gaussian CDF, logistic smoother, finite sums, and squaring operations are differentiable on the stated domain. Their composition is therefore differentiable. □

Proposition 4.2 (Zero-temperature limit)**.** *Fix PIT values such that $U_{t,r} \neq q_g$ for every t, r, g. As $\tau \downarrow 0$,* (5) *converges to the squared grid discrepancy formed with the empirical PIT CDF.*

Proof. For nonzero x, $\sigma(x/\tau) \to \mathbf{1}\{x > 0\}$ as $\tau \downarrow 0$. Every sum is finite, so the limit passes through the averages and squares. The exclusion of ties removes the ambiguous boundary value. □

Proposition 4.3 (Finite-grid interpretation)**.** *For fixed B, projections, grid, and $\tau > 0$, $\mathcal{L}_{\text{cal}}(\theta; B) = 0$ if and only if*

$$\frac{1}{|B|}\sum_{t\in B} s_\tau(q_g - U_{t,r}) = q_g \quad \textit{for every } (g, r).$$

Consequently, zero training loss establishes agreement only at the selected smoothed CDF constraints; it does not establish uniformity between grid points, for untested projections, conditional on the history, or out of sample.

Proof. Equation (5) is an average of nonnegative squared discrepancies. It is zero exactly when each squared discrepancy is zero. The stated limitations follow because the objective contains no other grid values, projections, conditioning events, or observations outside B. □

4.3 Sequential calibration loss

PIT uniformity does not rule out forecastable temporal structure. With $\widetilde{U}_{t,r} = U_{t,r} - 1/2$ and maximum lag K, define

$$\mathcal{L}_{\text{seq}}(\theta; B) = \frac{1}{RK} \sum_{r=1}^{R} \sum_{k=1}^{K} \left[\frac{1}{|B| - k} \sum_{t \in B : k \text{ available}} \widetilde{U}_{t,r} \widetilde{U}_{t-k,r} \right]^2 . \tag{6}$$

Under the ideal conditional forecast, the population quantities targeted by (6) are zero by the sequential PIT theorem. The converse is not true: finitely many zero autocovariances do not establish independence.

Proposition 4.4 (Exact scope of the sequential loss)**.** *Suppose $|B| > K$. Then $\mathcal{L}_{\text{seq}}(\theta; B) = 0$ if and only if*

$$\frac{1}{|B| - k} \sum_{t \in B : k \text{ available}} (U_{t,r} - 1/2)(U_{t-k,r} - 1/2) = 0 \quad \textit{for all } r = 1, \dots, R, \; k = 1, \dots, K.$$

If the forecast is correctly specified so that each projected sequential PIT is i.i.d. Unif$(0,1)$*, every corresponding population moment is zero. Conversely, satisfaction of these finitely many moments does not imply temporal independence, conditional uniformity, or correct joint dynamics.*

Proof. The first statement again follows because (6) is a sum of squares. Under i.i.d. uniform PITs, $\mathbb{E}(U_{t,r} - 1/2) = 0$ and independence gives $\mathbb{E}[(U_{t,r} - 1/2)(U_{t-k,r} - 1/2)] = 0$ for $k \geq 1$. The converse fails because a finite set of second-order moment restrictions cannot characterize an entire serial joint law; dependent processes can be uncorrelated at the tested lags. □

Remark 4.5. *The centered product in* (6) *uses the uniform-law mean* 1/2*, not the batch PIT mean. It is therefore a calibration moment rather than the usual sample autocovariance unless the batch mean is exactly* 1/2*.*

4.4 Frequentist CA-RNN objective

The proposed frequentist objective is

$$\widehat{\theta}_{\text{CA}} = \arg\min_{\theta} \left\{ \mathcal{L}_{\text{NLL}}(\theta; \mathcal{D}) + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}}(\theta; \mathcal{D}) + \lambda_{\text{seq}} \mathcal{L}_{\text{seq}}(\theta; \mathcal{D}) \right\} . \tag{7}$$

The replication experiment initially fixes $\lambda_{\text{seq}} = 0$, so that (7) follows the source paper's task-loss-plus-calibration structure. The sequential term is evaluated separately as a proposed extension.

4.5 Bayesian CA-RNN objective

Let every scalar neural parameter have variational posterior and prior

$$q_\phi(\theta_j) = \mathcal{N}(\mu_j, \sigma_j^2), \qquad p(\theta_j) = \mathcal{N}(0, \sigma_0^2), \qquad \sigma_j = \text{softplus}(\rho_j) + \epsilon.$$

Sampling uses the reparameterization

$$\theta_j = \mu_j + \sigma_j \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, 1).$$

The independent-Gaussian KL term has closed form

$$\mathrm{KL}(q_\phi \| p) = \sum_j \left[\log \frac{\sigma_0}{\sigma_j} + \frac{\sigma_j^2 + \mu_j^2}{2\sigma_0^2} - \frac{1}{2} \right]. \tag{8}$$

The calibration-aware variational objective is

$$\mathcal{F}^{\mathrm{CA}}(\phi) = \mathbb{E}_{\theta \sim q_\phi} \left[\mathcal{L}_{\mathrm{NLL}}(\theta; \mathcal{D}) + \lambda_{\mathrm{cal}} \mathcal{L}_{\mathrm{cal}}(\theta; \mathcal{D}) + \lambda_{\mathrm{seq}} \mathcal{L}_{\mathrm{seq}}(\theta; \mathcal{D}) \right] \tag{9}$$

$$+ \frac{\beta}{N} \mathrm{KL}(q_\phi(\theta) \| p(\theta)). \tag{10}$$

This is the sequential continuous-outcome analogue of the CA-BNN free energy in [1]. The factor $1/N$ matches the mean per-observation likelihood convention. The weight β remains a validation-selected generalized-Bayes hyperparameter.

Proposition 4.6 (Unbiased reparameterized gradient for the expectation)**.** *Assume differentiation and expectation may be interchanged. If $\theta^{(m)} = \mu + \sigma(\rho) \odot \varepsilon^{(m)}$ with independent $\varepsilon^{(m)} \sim \mathcal{N}(0, I)$, then the Monte Carlo gradient of the expected data-dependent part of* (10) *is unbiased.*

Proof. Write the expectation with respect to the parameter-free distribution of ε. Under the interchange assumption,

$$\nabla_\phi \mathbb{E}_\varepsilon [f(\mu + \sigma(\rho) \odot \varepsilon)] = \mathbb{E}_\varepsilon \nabla_\phi f(\mu + \sigma(\rho) \odot \varepsilon).$$

The sample average of independent terms has this expectation. The KL gradient is computed analytically from (8). □

Remark 4.7 (Dependent mini-batches)**.** *Unlike independent classification observations, overlapping time-series windows are dependent. Consecutive batches are required to evaluate $\mathcal{L}_{\mathrm{seq}}$, but their nonlinear calibration loss is not asserted to be an unbiased estimate of the full-sample calibration loss. Batch length and history overlap are therefore methodological hyperparameters and sensitivity targets.*

In the implementation, both calibration penalties are batch-local: lag pairs crossing mini-batch boundaries are omitted. Epoch summaries average batch means, so the final shorter batch receives the same summary weight as a full batch. The displayed dataset-level objectives are the target criteria; stochastic optimization uses this documented batch-local surrogate.

4.6 Why targeted calibration need not improve proper scores

This section gives a theoretical interpretation of the negative confirmatory result. The central distinction is between satisfying a finite collection of calibration moments and estimating the complete conditional distribution. The former is the explicit target of CA-RNN; the latter is assessed by strictly proper scores.

4.6.1 Population target and finite-projection non-identification

For a fixed forecast law Q and projection a_r, let $H_{Q,r}(u) = \mathbb{P}_Q(U_{t,r} \le u)$ denote the CDF of the resulting PIT under the true data law. The unsmoothed population counterpart of (5) on the implemented grid is

$$\mathcal{C}_{R,G}(Q) = \frac{1}{RG} \sum_{r=1}^{R} \sum_{g=1}^{G} \{H_{Q,r}(q_g) - q_g\}^2. \tag{11}$$

It measures unconditional agreement at RG points. It is not a metric on the space of joint conditional distributions.

Proposition 4.8 (Finite-grid calibration is non-identifying)**.** *Even in one dimension with a continuous strictly increasing true CDF F, there are infinitely many forecast laws $Q \neq P$ for which $\mathcal{C}_{1,G}(Q) = 0$ on any fixed finite grid.*

Proof. Choose any strictly increasing bijection $h : [0,1] \to [0,1]$ that fixes every grid point, $h(q_g) = q_g$, but is not the identity between at least two adjacent grid points. Define forecast law Q by its CDF $G(y) = h(F(y))$. If $Y \sim F$, then $G(Y) = h(F(Y))$ and, because $F(Y)$ is uniform,

$$\mathbb{P}\{G(Y) \leq q_g\} = h^{-1}(q_g) = q_g.$$

Thus every grid discrepancy is zero, while $G \neq F$. Infinitely many such monotone deformations h can be constructed between grid points. □

The multivariate case has two additional sources of non-identification: only a finite set of directions is inspected, and the moments are averaged across those directions. The Cramér–Wold device would identify a joint law if all one-dimensional projected distributions agreed, but it does not turn a finite projection set into an identifying criterion.

Proposition 4.9 (Consistency of fixed calibration moments)**.** *Fix Q, R, G, and $\tau > 0$. If the PIT process is stationary and ergodic, then the empirical smoothed grid moments converge almost surely to their population counterparts. The empirical lagged moments in* (6) *likewise converge almost surely for every fixed tested lag, provided their products are integrable.*

Proof. The smoothed indicators are bounded measurable functions of the PIT process, so Birkhoff's ergodic theorem applies to each sample average. The lagged products are measurable functions of the finite-dimensional shifted process and are integrable by assumption. There are finitely many projections, grid points, and lags, so convergence holds jointly for all implemented moments. □

This result concerns a fixed forecast rule under one stationary law. It does not provide uniform convergence over neural parameters, correct batch-local optimization bias, or transfer the moments to a shifted test law. Those missing implications are precisely relevant to the observed training-to-test gap.

4.6.2 Zero PIT autocovariance does not imply independence

The limitation of the sequential penalty is constructive, not merely technical.

Proposition 4.10 (Dependent uniform PITs with zero covariance)**.** *There exist dependent $U, V \sim$ $\mathrm{Unif}(0,1)$ satisfying $\mathbb{E}[(U - 1/2)(V - 1/2)] = 0$.*

Proof. Let $f(u) = 6u^2 - 6u + 1$ and define a bivariate density on $[0,1]^2$ by

$$c_\varepsilon(u, v) = 1 + \varepsilon f(u) f(v), \qquad 0 < |\varepsilon| < 1.$$

Because $\int_0^1 f(u)\, du = 0$, both marginals are uniform. The chosen range of ε keeps the density nonnegative. Since c_ε is not constant, U and V are dependent. However, symmetry of f about $1/2$ gives

$$\int_0^1 (u - 1/2) f(u)\, du = 0,$$

and hence

$$\mathbb{E}[(U-1/2)(V-1/2)] = \varepsilon \left\{ \int_0^1 (u-1/2)f(u)\,du \right\}^2 = 0.$$

□

Consequently, even population minimization of every moment represented by a finite-lag version of (6) cannot establish PIT independence. The confirmatory increase in PIT autocorrelation is stronger still: it shows that the estimated training moments did not generalize as lower corresponding test moments.

4.6.3 Energy Score regret is not controlled by calibration loss

For distributions with finite first moments, define

$$\mathrm{ES}(Q,y) = \mathbb{E}\|X-y\|_2 - \frac{1}{2}\mathbb{E}\|X-X'\|_2, \qquad X, X' \overset{\text{iid}}{\sim} Q.$$

Proposition 4.11 (Energy Score regret)**.** *If $Y \sim P$, then*

$$\mathbb{E}_P\,\mathrm{ES}(Q,Y) - \mathbb{E}_P\,\mathrm{ES}(P,Y) = \frac{1}{2}\mathcal{E}(P,Q) \geq 0, \tag{12}$$

where

$$\mathcal{E}(P,Q) = 2\mathbb{E}\|X-Y\|_2 - \mathbb{E}\|X-X'\|_2 - \mathbb{E}\|Y-Y'\|_2$$

is the energy distance. Equality holds only when $P = Q$.

Proof. Expand the two expected Energy Scores using independent $X, X' \sim Q$ and $Y, Y' \sim P$, then subtract. The remaining expression is one half of the energy distance. Its nonnegativity and identity of indiscernibles are the defining properties of energy distance on distributions with finite first moments. □

Equation (12) shows that Energy Score evaluates the complete joint distribution, whereas (11) evaluates only finitely many scalar constraints. No inequality in the CA-RNN objective bounds $\mathcal{E}(P,Q)$ by $\mathcal{C}_{R,G}(Q)$.

Proposition 4.12 (Calibration improvement need not improve a strictly proper score)**.** *For any fixed finite PIT grid, there exist a true continuous distribution P and forecasts Q_0, Q_1 such that*

$$\mathcal{C}_{1,G}(Q_1) < \mathcal{C}_{1,G}(Q_0), \qquad \mathbb{E}_P S(Q_1,Y) > \mathbb{E}_P S(Q_0,Y)$$

for any strictly proper score S that is finite and continuous along sufficiently small CDF perturbations of P.

Proof. Use the preceding monotone construction to choose $Q_1 \neq P$ with zero finite-grid calibration loss. Strict propriety gives a positive score regret for Q_1. Now choose $Q_0 \neq P$ as an arbitrarily small monotone perturbation of P that does not fix at least one grid point. Then $\mathcal{C}_{1,G}(Q_0) > 0$, while continuity makes its proper-score regret arbitrarily close to zero. Choose the perturbation small enough that its regret is below that of Q_1. The two displayed inequalities follow. □

This proposition formalizes the main empirical result: a lower projected-PIT error is not an ordering of forecast distributions and cannot, by itself, imply a lower Energy Score.

4.6.4 Why interval score can also worsen

Let F be the true marginal CDF, α the nominal miscoverage probability, $\tau = \alpha/2$, and $(\ell^*, u^*) = (F^{-1}(\tau), F^{-1}(1-\tau))$ the optimal central interval. For a candidate interval (ℓ, u), the expected interval-score regret has an exact quantile representation.

Proposition 4.13 (Interval-score regret)**.** *For continuous F with finite first moment,*

$$\mathbb{E}_F \operatorname{IS}_\alpha(\ell, u; Y) - \mathbb{E}_F \operatorname{IS}_\alpha(\ell^*, u^*; Y) \tag{13}$$

$$= \frac{1}{\tau}\left\{\int_{\ell^*}^{\ell} [F(x) - \tau]\, dx + \int_{u^*}^{u} [F(x) - (1-\tau)]\, dx\right\} \geq 0. \tag{14}$$

Proof. The interval score equals τ^{-1} times the sum of pinball losses at levels τ and $1-\tau$. The derivative of expected pinball loss with respect to a candidate quantile q is $F(q) - p$ at level p. Integrating this derivative from each true quantile to its candidate value yields (14). Monotonicity of F makes both oriented integrals nonnegative. □

The CA-RNN loss averages 19 grid levels and multiple directions and is optimized with smoothing on dependent training batches. A reduction in that average need not reduce coordinate-specific errors at the 0.05 and 0.95 quantiles on a shifted test distribution. Equation (14) shows that any such tail-quantile displacement creates positive expected interval-score regret. This explains how projected calibration can improve while the interval score worsens: the scalar calibration average falls, but errors at the particular marginal tail functionals priced by the interval score increase or the added width is not offset by sufficiently smaller miss penalties.

4.7 Training algorithms

For fixed-budget causal loss ablations, checkpoint selection and the patience test in Algorithm 1 are disabled, and all specifications are compared after the same number of epochs.

Algorithm 2 is the recurrent continuous-outcome adaptation of the CA-BNN procedure in [1].

5 Model Misspecification and Distribution Shift

5.1 Distinguishing misspecification from shift

Model misspecification occurs when the true conditional law $P_0(Y_{t+1} \mid \mathcal{F}_t)$ is outside the fitted family, even if the data law is stationary. Distribution shift occurs when the evaluation law differs from the development law. The two can coexist: a Gaussian predictive head is misspecified under heavy-tailed innovations, while a transition from Gaussian training innovations to heavy-tailed test innovations is also a distribution shift.

Let P_0 denote the reference law and let $P_{\delta,s}$ denote stress family s at severity $\delta \geq 0$, with $P_{0,s} = P_0$. For metric M, define degradation

$$\Delta_M^{(m,s)}(\delta) = M(P_{\delta,s}, \widehat{P}_m) - M(P_{0,s}, \widehat{P}_m), \tag{15}$$

where m indexes the forecasting method. For metrics in which smaller is better, positive Δ_M denotes deterioration. Coverage is reported as absolute error from the nominal level before differencing.

Algorithm 1 Frequentist CA-RNN training

Require: Ordered observations $\mathcal{D}$; history H; projections A; weights $\lambda_{\text{cal}}, \lambda_{\text{seq}}$; epoch limit E; gradient threshold c

Ensure: Fitted parameters $\widehat{\theta}$ selected without test information

Split $\mathcal{D}$ chronologically into $\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{te}}$
Fit standardization on $\mathcal{D}_{\text{tr}}$ and construct length-H windows
Initialize θ and fix $A = (a_1, \ldots, a_R)^{\mathsf{T}}$
for $e = 1, \ldots, E$ **do**
  for consecutive mini-batch $B \subset \mathcal{D}_{\text{tr}}$ **do**
    $(\mu_{\theta,t}, L_{\theta,t})_{t \in B} \leftarrow \textsc{CARNNForward}(B, \theta)$
    Compute $\mathcal{L}_{\text{NLL}}$ by (3) and projected PITs by (4)
    Compute $\mathcal{L}_{\text{cal}}$ by (5) and $\mathcal{L}_{\text{seq}}$ by (6)
    $\mathcal{J} \leftarrow \mathcal{L}_{\text{NLL}} + \lambda_{\text{cal}}\mathcal{L}_{\text{cal}} + \lambda_{\text{seq}}\mathcal{L}_{\text{seq}}$
    $g \leftarrow \textsc{ClipGradNorm}(\nabla_\theta \mathcal{J}, c)$
    Update θ with AdamW using g
  end for
  Evaluate the prespecified selection score on $\mathcal{D}_{\text{val}}$
  Save θ if the validation score improves
  if the patience criterion is met **then**
    break
  end if
end for
Restore the selected checkpoint $\widehat{\theta}$
Evaluate $\widehat{\theta}$ once on $\mathcal{D}_{\text{te}}$

5.2 Stress families

The following are prospective isolated stress families for a confirmatory study; they were not all executed factorially in the present experiments:

1. **Scale shift:** innovation covariance is multiplied by a severity function while dependence is held fixed.

2. **Dependence shift:** correlations or a dynamic correlation state change while marginal variances are controlled.

3. **Tail shift:** Gaussian innovations transition to Student-t, mixtures, or asymmetric downside contamination.

4. **Volatility misspecification:** ARCH/GARCH or stochastic-volatility dynamics violate the Gaussian head's conditionally homoskedastic approximation.

5. **Mean-dynamics shift:** nonlinear interactions or an abrupt change in recurrent coefficients alter the conditional location.

The implemented developmental curve uses common random numbers and jointly changes scale and common correlation. Models are selected without examining its stressed test metrics and are frozen after fitting at $\delta = 0$. Tail, volatility, and mean-dynamics regimes appear in separate developmental simulations, not in a completed isolated-factor robustness experiment.

Algorithm 2 Variational calibration-aware Bayesian RNN training

Require: Ordered observations $\mathcal{D}$; projections A; prior scale σ_0; Monte Carlo count M; loss weights $\lambda_{\text{cal}}, \lambda_{\text{seq}}, \beta$; epoch limit E

Ensure: Variational parameters $\widehat{\phi} = (\widehat{\mu}, \widehat{\rho})$

```
Construct chronological splits and training-only standardization as in Algorithm 1
Initialize φ = (μ, ρ) and set σ(ρ) = softplus(ρ) + ϵ
for e = 1, ..., E do
    for consecutive mini-batch B ⊂ D_tr do
        for m = 1, ..., M do
            Draw ε^(m) ~ N(0, I) and set θ^(m) ← μ + σ(ρ) ⊙ ε^(m)
            Use θ^(m) consistently at every recurrent time step
            Compute J_data^(m) ← L_NLL^(m) + λ_cal L_cal^(m) + λ_seq L_seq^(m)
        end for
        F̂^CA ← M^{-1} Σ_{m=1}^{M} J_data^(m) + (β/N) KL(q_φ ‖ p)
        Differentiate F̂^CA, clip its gradient, and update φ with AdamW
    end for
    Evaluate validation NLL at posterior-mean weights and update the selected checkpoint
    if the patience criterion is met then
        break
    end if
end for
Restore φ̂
for each test origin t and predictive draw s do
    Draw θ^(s) ~ q_φ̂ and then Y_{t+1}^(s) ~ p_{θ^(s)}(· | F_t)
end for
Evaluate the posterior predictive mixture on D_te
```

5.3 Robustness estimands

For paired realization b, method m, comparator c, and metric M, define

$$D_b^{m,c}(\delta) = M_b^m(\delta) - M_b^c(\delta).$$

For a future experiment with B independent DGP realizations, the intended summary is

$$\overline{D}^{m,c}(\delta) = \frac{1}{B}\sum_{b=1}^{B} D_b^{m,c}(\delta), \qquad \operatorname{MCSE}(\overline{D}) = \frac{s_D(\delta)}{\sqrt{B}}.$$

Neural-seed repetitions on a prespecified subset separate DGP variability from optimization variability. Failure frequencies and runtimes are part of the estimand rather than discarded implementation details.

6 Experimental Design

6.1 Four-model replication matrix

The core experiment follows the comparison logic of [1]:

Table 1: Mapping from calibration-aware classification to forecasting.

Source-paper model	Forecasting model	Objective components
FNN	RNN	NLL
BNN	BRNN	expected NLL + KL
CA-FNN	CA-RNN	NLL + projected-PIT calibration
CA-BNN	CA-BRNN	expected NLL/calibration + KL

Architectures, histories, splits, optimizers, stopping rules, and predictive heads are matched wherever the Bayesian formulation permits. The same projection matrix is used for all models in a comparison.

6.2 Conventional benchmark ladder

The neural comparisons are supplemented by three increasingly dynamic probabilistic benchmarks. A centered historical bootstrap samples joint training returns and contains neither a dynamic conditional mean nor time-varying scale. A Gaussian VAR estimates a linear conditional mean and a fixed innovation covariance. The strongest conventional benchmark combines that VAR mean with separate marginal GARCH$(1,1)$ filters,

$$h_{j,t} = \omega_j + \alpha_j e_{j,t-1}^2 + \beta_j h_{j,t-1}, \qquad \omega_j > 0, \quad \alpha_j, \beta_j \geq 0, \quad \alpha_j + \beta_j < 1,$$

and samples entire standardized-residual vectors $z_t = (e_{1,t}/\sqrt{h_{1,t}}, \ldots, e_{d,t}/\sqrt{h_{d,t}})$ from the training period. Its one-step samples are

$$Y_t^{(s)} = \widehat{\mu}_t + \mathrm{diag}(\sqrt{\widehat{h}_t}) z_{I_s},$$

where I_s is a common row index across margins. This preserves empirical contemporaneous dependence and tail shape while allowing marginal conditional volatility to evolve. The VAR lag is selected from $\{1, 5, 10, 20\}$ by validation one-step MSE; parameters and the residual resampling distribution use training data only.

Model checking for the selected benchmark is also training-only. VAR dynamic stability is assessed by the spectral radius of its companion matrix. GARCH admissibility is checked through optimizer convergence, nonnegative parameters, $\alpha_j + \beta_j < 1$, and the frequency with which filtered variances reach the numerical floor. Ljung–Box statistics at lags 5, 10, and 20 are computed for standardized residuals and their squares to diagnose remaining conditional-mean and volatility dependence. Standardized-residual skewness, kurtosis, Jarque–Bera tests, and the residual correlation matrix assess the rationale for resampling empirical residual vectors jointly. The Ljung–Box and Jarque–Bera p-values are descriptive because they do not account for VAR–GARCH parameter estimation or multiplicity.

6.3 Frozen external-panel replication

The confirmatory protocol is stored separately in `docs/confirmatory_protocol.md` before outcome inspection. It fixes a disjoint MSFT–BAC–CVX–COST panel, the date boundary, transformations, split, architecture, loss weights, training and forecast seeds, primary outcomes, baseline definitions, bootstrap procedure, and failure reporting. The sole primary contrast is sequential CA-RNN minus RNN across ten matched optimization seeds; conventional-baseline comparisons are secondary. Any post-outcome change creates an exploratory analysis rather than replacing the frozen result.

6.4 Calibration-weight experiment

Analogously to the source paper's plot of ECE and accuracy against λ, this study plots projected-PIT calibration error and predictive performance against λ_{cal}. The paired panels will contain:

1. projected calibration discrepancy and reliability curves;
2. Energy Score, NLL where comparable, RMSE, interval score, and width.

The result of interest is a Pareto relationship, not calibration alone. Separate curves are required for CA-RNN and CA-BRNN.

6.5 Sequential ablation

The minimum ablation is: The Bayesian versions repeat this comparison with the KL term always

Table 2: Loss-component ablation.

Specification	NLL	$\mathcal{L}_{\text{cal}}$	$\mathcal{L}_{\text{seq}}$
RNN	✓		
CA-RNN	✓	✓	
CA-RNN (sequential)	✓	✓	✓

present. Additional ablations vary coordinate-only versus augmented projections, smoothing temperature, batch length, lag order, prior scale, and β.

To prevent checkpoint selection from masquerading as a calibration effect, the primary loss-component ablation uses identical initial parameters, chronological mini-batches, optimization budgets, and predictive random numbers, with early stopping disabled. A secondary operational comparison may use early stopping, but all methods must then be selected by the same validation NLL rule.

6.6 Controlled simulations

Development simulations include Gaussian, heavy-tailed, heteroskedastic, and nonlinear regimes. A future confirmatory experiment should use independently generated data for every regime, sample size, shift severity, and DGP seed, with the following fixed in advance:

- sample sizes and number of independent realizations;
- neural-seed subset and number of optimization repetitions;
- calibration and KL grids;
- projection directions and smoothing grid;
- primary metrics and failure definitions; and
- rules for early stopping and invalid numerical runs.

6.7 Financial application

The application uses adjusted daily prices for AAPL, JPM, XOM, and WMT beginning in 2007 and converts them to aligned log returns. The four assets provide a small, computationally manageable multivariate system with heterogeneous sector exposure. All splits are chronological. Development selected $\lambda_{\text{cal}} = 100$ and $\lambda_{\text{seq}} = 2000$ before the multi-seed and rolling-regime analyses. The focused frequentist comparison uses a 32-unit GRU, a 20-day history, a joint Gaussian head, 500 forecast draws, and validation-NLL early stopping.

Standardization is fitted on each training partition. Validation contexts carry the final 20 training observations, and test contexts carry the final 20 validation observations; every evaluation target therefore retains an available historical context without using future information. RMSE, Energy Score, interval width, and interval score are reported in training-standardized units. Coverage and PIT diagnostics are invariant to the componentwise affine transformation. The two penalty weights were chosen after inspecting developmental simulation and ablation outcomes, rather than by a pristine validation-only or preregistered selection procedure.

The rolling analysis uses expanding estimation samples and four non-overlapping test periods: 2017–2019 (pre-pandemic), 2020–2021 (pandemic), 2022–2023 (inflation and monetary tightening), and 2024–2025 (recent). Each window has a strictly subsequent validation period and five matched optimization seeds. These labels describe calendar periods rather than uniquely identified causal regimes. Because the full panel was observed during development and the final two windows overlap the initial test period, this is temporal robustness evidence rather than a pristine preregistered confirmatory experiment.

6.8 Status of the evidence

The lambda sweep, ablation, original-panel ten-seed study, rolling windows, and synthetic stress curve are developmental or internal-validation evidence. They condition on a specification informed by earlier results. The external-panel replication is confirmatory for its single primary contrast: sequential CA-RNN minus RNN on MSFT, BAC, CVX, and COST under the protocol frozen in `docs/confirmatory_protocol.md`. Its traditional-baseline comparisons were prespecified but secondary. Inferential intervals quantify test-origin and optimization-seed uncertainty under the implemented design; they do not include uncertainty over asset populations, architectures, or alternative research decisions.

6.9 Paired uncertainty design

For model m, comparator c, seed s, and forecast origin t, let $d_{s,t}^{m,c}$ be a paired additive-score difference. The estimand is

$$\widehat{\Delta}^{m,c} = \frac{1}{S}\sum_{s=1}^{S}\frac{1}{T}\sum_{t=1}^{T} d_{s,t}^{m,c}.$$

The hierarchical bootstrap resamples optimization seeds and, within each selected seed, matched forecast origins in moving blocks of 20 trading days. Projected-PIT calibration, coverage error, and PIT dependence are recomputed in every bootstrap draw. The initial financial analysis uses 2,000 draws and each rolling regime uses 1,000. Raw coverage and width remain descriptive; the inferential coverage loss is

$$E_{\text{cov}} = |\widehat{C}_{0.90} - 0.90|.$$

Negative candidate-minus-comparator differences favor the candidate for loss metrics. An interval containing zero is called unresolved, not evidence of equality. Inference conditions on the architecture, tuning path, projection set, assets, and observed calendar windows.

6.10 Metrics and figures

The tables report mean performance, paired differences, and uncertainty intervals. The diagnostic figures include:

1. calibration and predictive metrics versus λ_{cal};
2. nominal-versus-empirical coverage curves;
3. projected PIT histograms or CDF reliability curves;
4. PIT autocorrelation diagnostics;
5. degradation curves versus shift severity; and
6. a calibration–sharpness or calibration–Energy Score frontier.

7 Results

7.1 Implementation validation

The implementation tests cover tensor dimensions, covariance positive-definiteness, loss evaluation, forecasting, chronological data handling, traditional baselines, and paired-bootstrap identities. Twenty-three automated tests pass. The notebooks also completed the simulation, shift, financial, optimization-seed, and rolling-regime workflows. Successful execution establishes implementation consistency but is not itself statistical validation.

7.2 Four-model comparison

The initial matched comparison showed that calibration-aware objectives alter the forecast distribution but do not yield uniform dominance. CA-RNN made only a small projected-calibration improvement over RNN, while CA-BRNN did not improve on BRNN. Bayesian averaging did not remove the calibration–score tradeoff. The frequentist RNN, CA-RNN, and sequential CA-RNN therefore form the focused empirical comparison; BRNN and CA-BRNN remain supporting replications of the source paper's model matrix.

7.3 Calibration-weight sensitivity

The exploratory sweep over $\lambda_{\text{cal}} \in \{0, 1, 10, 50, 100, 250, 500\}$ displayed a Pareto tradeoff: projected calibration improved through the intermediate penalty range, whereas likelihood and proper-score performance deteriorated. The value 100 was retained as a developmental compromise rather than selected to maximize a final test result. Calibration improvement alone is therefore not interpreted as a better probabilistic forecast.

7.4 Sequential calibration ablation

Relative to marginal CA-RNN, the sequential specification consistently repaired predictive performance. In the ten-seed financial experiment it improved MSE, Energy Score, interval score, nominal-coverage error, and projected-PIT calibration for every seed. It widened intervals, but simultaneous coverage and interval-score improvements show that the result is not merely unpenalized widening. Improvements in PIT temporal dependence were smaller and generally unresolved. The sequential penalty is empirically important, but the evidence does not establish PIT independence.

7.5 Misspecification and distribution shift

The frozen-model experiment jointly increased innovation scale and common correlation from severity zero to one. All specifications deteriorated. At zero, RNN had Energy Score 1.2022, interval score 3.8595, and projected calibration error 0.000866, compared with 1.2039, 3.9311, and 0.001345 for sequential CA-RNN. At maximum severity, the ordering reversed: RNN obtained 2.3789, 10.5168, and 0.009528, while sequential CA-RNN obtained 2.3474, 10.0759, and 0.007717. This single-seed combined stress is developmental evidence rather than an isolated causal test of scale or dependence. Absolute levels and paired degradation are considered together because a worse nominal starting point can mechanically reduce measured deterioration.

7.6 Financial results

Table 3 reports averages over ten matched optimization seeds and 957 forecast origins on the original financial test period. Sequential CA-RNN reduced projected-PIT calibration error relative to RNN for all ten seeds, from 0.001438 to 0.001225, or approximately 15%. Its Energy Score and RMSE were worse for all ten seeds. Its mean interval score was slightly lower, but the paired difference remained unresolved, with a 95% hierarchical interval of $[-0.0191, 0.0136]$. In contrast, its coverage-error difference was -0.00528 with interval $[-0.00815, -0.00136]$, favoring sequential CA-RNN.

Table 3: Mean financial performance over ten matched optimization seeds. Lower is better except for coverage, whose nominal target is 0.90.

Model	RMSE	Energy	Interval	Coverage	PIT calibration
RNN	0.9399	1.1450	4.0855	0.8717	0.001438
CA-RNN	0.9504	1.1563	4.1191	0.8666	0.001306
Sequential CA-RNN	0.9479	1.1531	4.0816	0.8770	0.001225

Plain CA-RNN isolates why the extension matters. Against RNN it had worse MSE, Energy Score, interval score, and coverage error despite lower projected-PIT error. Sequential CA-RNN then reduced MSE by 0.00465, Energy Score by 0.00317, interval score by 0.03749, and coverage error by 0.01032 relative to CA-RNN; every corresponding hierarchical interval excluded zero. It repairs a substantial part, but not all, of the cost of marginal calibration.

7.7 Conventional-baseline results

Table 4 adds the strongest traditional benchmark to the original panel. Validation selected VAR lag one, and all four marginal GARCH optimizations converged. The fixed-covariance Gaussian VAR was poorly calibrated; adding marginal GARCH scales and resampling joint standardized residuals reduced its interval score from 4.3993 to 4.0298 and projected-PIT error from 0.003388 to 0.000453. VAR–GARCH had better interval score, coverage error, and projected-PIT calibration than sequential CA-RNN. It also had slightly better RMSE and Energy Score. The comparison attributes much of the neural models' remaining uncertainty error to unmodeled conditional volatility rather than an insufficiently flexible conditional mean alone.

Table 4: Original-panel neural and VAR–GARCH results. Neural entries are ten-seed means; the traditional model is fitted once. Coverage targets 0.90.

Model	RMSE	Energy	Interval	Coverage	PIT calibration
RNN	0.9399	1.1450	4.0855	0.8717	0.001438
Sequential CA-RNN	0.9479	1.1531	4.0816	0.8770	0.001225
VAR–GARCH bootstrap	0.9452	1.1495	**4.0298**	**0.8989**	**0.000453**

Against sequential CA-RNN, the paired hierarchical intervals favored VAR–GARCH for MSE, Energy Score, interval score, coverage error, and projected-PIT calibration. These intervals repeat the fixed traditional forecast across neural seed slots; they represent test-origin dependence and neural optimization variation but condition on the fitted baseline.

The training-sample specification checks in Table 5 support the numerical and volatility adequacy of the benchmark. The VAR companion spectral radius was 0.1513, well below one. Every GARCH optimization converged, all persistence estimates were below one, and no filtered variance reached the numerical floor. Persistence was nevertheless high, ranging from 0.9700 to 0.9894, so volatility shocks decay slowly. None of the 12 Ljung–Box checks on squared standardized residuals rejected at the 5% level, providing no evidence of remaining volatility dependence at the tested lags.

Table 5: Training-sample diagnostics for the selected VAR(1)–GARCH benchmark. $p_{z,20}$ and $p_{z^2,20}$ are descriptive Ljung–Box p-values at lag 20; kurtosis is the ordinary, rather than excess, standardized-residual kurtosis.

Asset	$\alpha + \beta$	$p_{z,20}$	$p_{z^2,20}$	Kurtosis
AAPL	0.9700	0.2307	0.9689	6.30
JPM	0.9849	0.1490	0.7370	5.53
XOM	0.9849	0.0142	0.8253	4.88
WMT	0.9894	0.0089	0.8922	11.54

The conditional-mean diagnostics are less favorable. Six of the 12 Ljung–Box tests on standardized residuals rejected at 5%; XOM rejected at all three tested lags, WMT at lags 10 and 20, and JPM at lag 5. Thus VAR(1) leaves some linear serial dependence even though validation preferred it to the longer-lag candidates. All four Jarque–Bera tests strongly rejected Gaussian standardized innovations, and marginal kurtosis ranged from 4.88 to 11.54. Pairwise standardized-residual correlations ranged from 0.22 to 0.49. The heavy tails and cross-sectional dependence jointly support the use of synchronized empirical residual vectors rather than Gaussian or independently resampled innovations. Accordingly, VAR–GARCH is treated as a strong volatility-aware benchmark, not a fully specified model of the conditional mean.

7.8 Frozen external-panel replication

The SHA-256 hash recorded with the executed results matches the frozen protocol. The replication contains ten paired optimization seeds and 989 test origins; all marginal GARCH optimizations converged. Table 6 reports model means.

For the prespecified primary contrast, sequential CA-RNN reduced projected-PIT calibration error by 0.0000473, with 95% interval $[-0.000163, -0.000012]$. This targeted effect therefore replicated on disjoint assets. It did not yield a better predictive distribution overall: candidate-

Table 6: Frozen MSFT–BAC–CVX–COST replication. Neural entries are ten-seed means; coverage targets 0.90.

Model	RMSE	Energy	Interval	Coverage	PIT cal.	PIT dep.
RNN	0.8609	1.0677	3.7766	0.8703	0.000783	0.03008
Sequential CA-RNN	0.8694	1.0778	3.7998	0.8683	0.000735	0.03378
VAR–GARCH bootstrap	0.8646	1.0713	3.7528	0.8928	0.000491	0.03270

minus-RNN differences were +0.01476 for MSE, +0.01007 for Energy Score, and +0.02318 for interval score, with respective intervals $[0.01236, 0.01751]$, $[0.00847, 0.01189]$, and $[0.00773, 0.03937]$. The coverage-error difference was unresolved. Projected PIT mean absolute autocorrelation was also worse by 0.00370, with interval $[0.00005, 0.00455]$. Thus the sequential objective did not improve out-of-sample temporal PIT adequacy despite targeting finite-lag moments during training.

The secondary comparison again favored VAR–GARCH over sequential CA-RNN for MSE, Energy Score, interval score, coverage error, and projected calibration; all five intervals excluded zero. The RNN retained better MSE and Energy Score than VAR–GARCH, whereas VAR–GARCH had better interval score, coverage error, and projected calibration. Conditional-mean accuracy and uncertainty calibration are therefore supplied by different model components in this application.

7.9 Rolling market-regime results

Table 7 gives five-seed means. Sequential CA-RNN had lower projected-PIT calibration error than RNN in all four regimes. Hierarchical intervals favored it in the pre-pandemic, inflation/tightening, and recent windows; the pandemic difference was unresolved. In contrast, its Energy Score was lower only pre-pandemic, where the difference was unresolved.

Table 7: Five-seed regime means for RNN (R) and sequential CA-RNN (S).

	Energy Score		Interval score		PIT calibration	
Regime	R	S	R	S	R	S
Pre-pandemic	0.8629	0.8626	3.1123	3.1027	0.001914	0.001503
Pandemic	1.5016	1.5029	5.7371	5.6629	0.002995	0.002883
Inflation/tightening	1.2809	1.2884	4.5862	4.5396	0.001842	0.001465
Recent	1.0237	1.0289	3.6648	3.6935	0.001224	0.000904

The interval-score ordering is regime-dependent. Sequential CA-RNN significantly outperformed RNN during the pandemic and inflation/tightening periods, was unresolved pre-pandemic, and significantly underperformed recently. Coverage error similarly favored sequential CA-RNN pre-pandemic and during inflation/tightening, was unresolved during the pandemic, and favored RNN recently. These reversals directly answer the distribution-shift question.

Across regimes, sequential CA-RNN improved Energy Score and interval score over marginal CA-RNN directionally in four of four windows. These improvements were resolved in the pandemic, inflation/tightening, and recent windows, while the pre-pandemic differences were unresolved. Projected calibration improved over marginal CA-RNN in three of four windows.

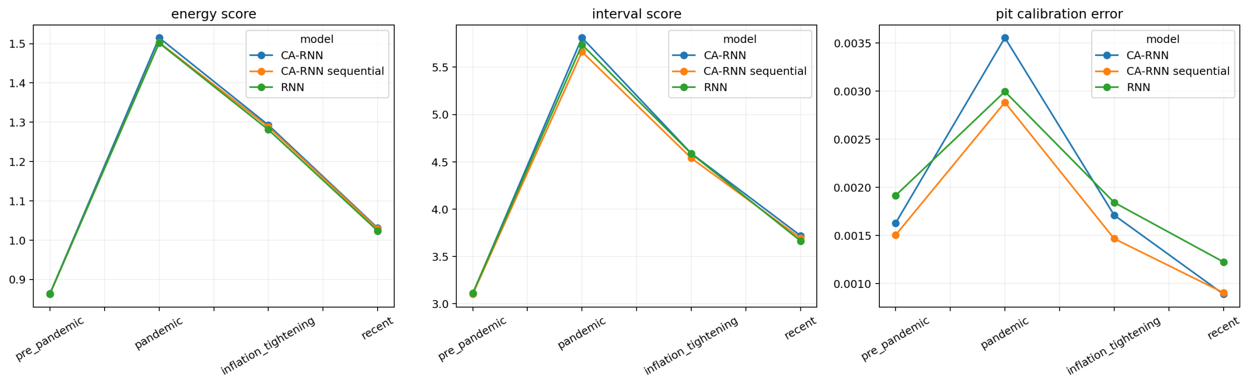

Figure 1: Mean Energy Score, interval score, and projected-PIT calibration error across five matched seeds and four chronological test regimes. Lower is better.

7.10 Interval-score mechanism under temporal shift

To explain the regime reversal, the retained forecasts were analyzed without refitting. For a 90% interval $[L_{t,j}, U_{t,j}]$, the componentwise interval score decomposes exactly as

$$S_{t,j} = \underbrace{U_{t,j} - L_{t,j}}_{\text{width}} + \underbrace{20(L_{t,j} - Y_{t,j})\mathbf{1}\{Y_{t,j} < L_{t,j}\}}_{\text{lower-tail penalty}} + \underbrace{20(Y_{t,j} - U_{t,j})\mathbf{1}\{Y_{t,j} > U_{t,j}\}}_{\text{upper-tail penalty}}. \tag{16}$$

Table 8 reports the mean sequential-CA-RNN-minus-RNN difference in each component. The arithmetic identity in (16) means the final column is exactly the sum of the preceding three columns.

Table 8: Decomposition of the interval-score difference between sequential CA-RNN and RNN. Negative total differences favor sequential CA-RNN.

Regime	Width	Lower penalty	Upper penalty	Total
Pre-pandemic	−0.0439	+0.0241	+0.0103	−0.0096
Pandemic	+0.0420	−0.0599	−0.0563	−0.0742
Inflation/tightening	+0.2342	−0.1709	−0.1098	−0.0466
Recent	−0.0185	+0.0268	+0.0205	+0.0287

The pandemic and inflation/tightening improvements arise through beneficial widening: the added width is more than offset by reductions in both tail-miss penalties. In the recent period, sequential CA-RNN instead produces slightly narrower intervals and larger penalties in both tails. Before the pandemic, the mechanism is different again: reduced width outweighs the additional tail penalties, yielding a small and statistically unresolved sharpness benefit.

The stress-period effect is concentrated in forecasts preceded by elevated volatility. Sequential CA-RNN's interval-score differences in the pandemic are +0.0198, −0.0725, and −0.1705 for low-, medium-, and high-volatility origins, respectively. During inflation/tightening they are +0.0209, −0.0562, and −0.1045. By contrast, the recent-period differences are positive in every state: +0.0318, +0.0238, and +0.0305. Volatility states use trailing 20-day realized volatility known before each forecast and within-regime terciles.

The improvements are also asset-concentrated. XOM accounts for most of the inflation/tightening gain, with an interval-score difference of −0.2364; AAPL, JPM, and WMT have differences of −0.0051, +0.0221, and +0.0331. Pandemic improvements occur for AAPL, XOM, and WMT but not JPM. Recent losses are concentrated in AAPL, XOM, and WMT, while JPM improves slightly. These results preclude interpreting an aggregate gain as uniform across assets.

The train-to-test descriptors support, but do not prove, a scale-and-tail interpretation. Pandemic volatility is 1.26 times its training level and the mean standardized fourth moment is 29.87. The corresponding values are 1.00 and 12.08 during inflation/tightening, versus 0.83 and 6.37 recently. The recent period has the largest standardized location shift and correlation shift of the four windows. Thus sequential calibration appears most useful for persistent scale and tail errors, while lower-volatility location or dependence changes remain harder.

Importantly, projected and coordinate PIT calibration still improve recently even as 90% interval performance deteriorates. The calibration loss averages 19 CDF grid levels and multiple projections; it does not target the 5% and 95% quantiles specifically. Better global PIT uniformity can therefore coexist with worse tail coverage. The exact score decomposition is established by the retained forecasts, whereas the proposed economic mechanism is post-hoc and requires new data for confirmation.

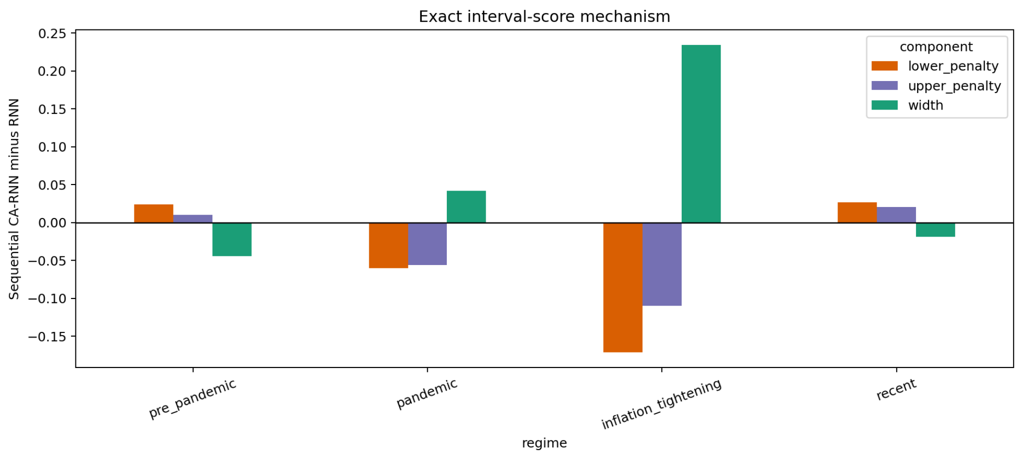

Figure 2: Exact decomposition of sequential-CA-RNN-minus-RNN interval-score differences. Tail reductions offset widening in the pandemic and inflation/tightening periods, but not in the recent period.

8 Discussion

8.1 Interpretation

Four conclusions follow. First, a marginal calibration penalty is insufficient for dependent multivariate forecasting: it can improve its targeted diagnostic while harming proper scores and nominal coverage. Second, the sequential loss is not cosmetic within the developmental ablation; it improves marginal CA-RNN and recovers part of its predictive-performance loss. Third, its projected-calibration effect transfers to a disjoint asset panel, but its developmental interval and coverage improvements do not: the frozen replication instead favors RNN on all three proper or accuracy-oriented scores and finds no resolved coverage benefit. Fourth, VAR–GARCH shows that explicit volatility dynamics and joint residual resampling can provide better-calibrated uncertainty than direct calibration regularization in these data.

The disagreement between Energy Score and interval score is substantively useful. Energy Score assesses the multivariate predictive distribution, whereas the marginal interval score rewards coverage and sharpness asset by asset. During the pandemic and inflation/tightening periods, sequential CA-RNN improves the latter without improving the former. Its practical value therefore depends on whether an application prioritizes marginal risk intervals or overall multivariate accuracy. The confirmatory evidence narrows this claim further: lower projected-PIT grid error alone is insufficient evidence of improved operational uncertainty.

The regret identities explain the observed divergence. Energy Score regret is one half of the energy distance and therefore responds to discrepancies anywhere in the joint distribution. Marginal interval-score regret is the sum of two integrated quantile errors at the selected tail probabilities. By contrast, the training regularizer averages a finite set of smoothed PIT-CDF moments across grid points and projections. Because this finite criterion is non-identifying, reducing it supplies no mathematical upper bound on either regret. The external result—lower projected-PIT error alongside higher Energy and interval scores—is therefore compatible with the objectives rather than paradoxical.

8.2 Limitations

Limitations include finite projection sets, two small four-asset panels, one-step horizons, sensitivity to smoothing and batching, mean-field variational misspecification, a Gaussian predictive head, and limited ability to establish conditional calibration from finite dependent sequences. Regime labels are calendar-based and partly specified after broader exposure to the data. The hierarchical bootstrap represents seed and within-window sampling variation but conditions on the architecture, tuning path, assets, projections, and chosen windows. Conventional baselines are fitted once per panel, so their reported hierarchical comparisons do not represent baseline parameter-estimation variability. Moving-block resampling can distort PIT autocorrelation at artificial block boundaries, so temporal-dependence intervals are interpreted cautiously. Finally, improved finite-projection calibration does not imply full joint or conditional calibration. The original-panel VAR–GARCH diagnostics are also in-sample and descriptive: several standardized-residual Ljung–Box tests reject, and their p-values are not adjusted for fitted parameters or multiple testing. Its comparative advantage should therefore be attributed primarily to effective conditional-volatility filtering and joint empirical innovation resampling, not to complete dynamic specification.

8.3 Future work

The most direct methodological extension is a tail-aware sequential calibration objective. The current loss weights all 19 PIT grid levels and all fixed projections equally. A weighted version could emphasize application-relevant tail levels,

$$\mathcal{L}_{\text{tail}} = \frac{1}{R\sum_g w_g}\sum_{r=1}^{R}\sum_{g=1}^{G} w_g \left[\frac{1}{|B|}\sum_{t\in B} s_\tau(q_g - U_{t,r}) - q_g\right]^2, \tag{17}$$

with larger prespecified w_g near $q = 0.05$ and $q = 0.95$. A complementary extension would penalize lagged dependence in tail exceedance indicators rather than only covariance of centered PITs. Either extension must be tuned on validation data and evaluated with global calibration and proper scores to avoid improving one nominal interval at the expense of the rest of the distribution.

Further work should test new frequencies, forecast horizons, broader asset universes, and genuinely future calendar periods. Conditional calibration diagnostics could use volatility, market return, and liquidity states fixed before evaluation. Heavy-tailed or mixture predictive heads may separate deficiencies in the Gaussian output law from deficiencies in the calibration objective.

Finally, uncertainty for PIT dependence should use a procedure that avoids correlations introduced at moving-block boundaries.

9 Conclusion

This paper adapts calibration-aware Bayesian learning to continuous multivariate recurrent forecasting and introduces a temporal calibration penalty based on lagged projected-PIT covariance. Calibration-aware training reliably changes forecast calibration, but marginal regularization alone incurs predictive costs. The sequential extension improves the marginal CA-RNN specification in the developmental ablation and produces lower projected-PIT calibration error than a likelihood-trained RNN across optimization seeds, chronological regimes, and a frozen disjoint-asset replication.

The method is not uniformly superior. In the confirmatory experiment it has worse MSE, Energy Score, interval score, and temporal PIT dependence than RNN, while its coverage-error difference is unresolved. VAR–GARCH also outperforms it on the principal uncertainty metrics in both asset panels. The contribution is therefore not a dominance claim, nor evidence that the present sequential penalty achieves PIT independence. It is a controlled demonstration that calibration-aware training can reproducibly improve its targeted finite-projection diagnostic while failing to improve—and sometimes worsening—proper scores and sequential adequacy. Strong volatility-aware baselines and external confirmation are thus essential when evaluating calibration-aware neural forecasts.

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